Expense Mate

Members: Areesha 24k 3043, Burhan 24k 3056

Phase 05

Maintenance, Evolution, Refactoring and Lehman's Laws Report

Deliverable:	Change Log + Refactoring Report (Phase 5)
Authors:	**Areesha** (Corrective Maintenance & Complexity Metrics) & **Burhan** (Adaptive & Perfective Maintenance)
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152 of 152 tests passing (100%) · 97% code coverage · average cyclomatic complexity 3.01, Grade A · maintainability Grade A across every module

1. What Happened in Phase 5

Phase 4 left ExpenseMate at an integrated baseline. Phase 5 moved it into active maintenance and refactoring, following the ISO/IEC 14764 and IEEE 1219 standards for how this kind of work is classified and documented. The team did four things:

- **Corrective (Areesha):** fixed two real bugs. One caused badly overspent categories to still show a Within Budget status, and the other caused negative amounts to display incorrectly because of how Python handles division.
- **Adaptive (both members):** added the ability to export CSVs by date range so exports match how people actually do their accounting, covering all transactions, the current month, or a custom range.
- **Perfective (Burhan):** added a multi select checklist for deleting transactions in bulk, modeled on a typical email inbox, including a Select All option and a guarded option to delete everything.
- **Evolution analysis (both members):** a short review of ExpenseMate against all eight of Lehman's Laws of Software Evolution, a framework describing how software systems need to keep changing to remain useful.

2. Change Log

ID	Date	Author	Type	Modules Touched	What Changed and Why
CHG-01	2026 09 25	Areesha	**Corrective**	`budget_calculator.py`, `ui/components.py`	Fixes BUG-09: the over budget check and status display now agree, so overspent categories correctly show Over Budget.
CHG-02	2026 09 25	Areesha	**Corrective**	`services/money.py`	Fixes BUG-10: separates the sign from a negative amount before dividing, so amounts no longer round incorrectly.
CHG-03	2026 09 25	Both	**Adaptive**	`transaction_service.py`, `pages/5_Import_Export.py`	Adds date range filtering to CSV exports (all, current month, or custom), matching how people actually do their accounting.
CHG-04	2026 09 25	Burhan	**Perfective**	`transaction_repo.py`, `transaction_service.py`, `pages/1_Transactions.py`	Adds batch deletion support, plus an email style multi select checklist with Select All and a two part confirmation.
CHG-05	2026 09 25	Both	**Regression**	`tests/test_maintenance_phase5.py`	Added 8 new automated tests covering both bug fixes, scoped export, and bulk delete. Total test count is now 152.

3. Bug Fixes (Corrective Maintenance)

BUG-09: Categories Showed Within Budget While Badly Overspent

- **Severity:** High

- **Where:** Budget calculation logic and the status display component.
- **What happened:** During manual testing, categories that were severely overspent, in one case close to five times the budget, still displayed a status of Within Budget instead of Over Budget.
- **Cause:** The over budget check used flawed logic and returned an incorrect result once spending passed the limit, and that incorrect result carried through to the status shown on screen.
- **Fix:** The check now strictly compares spending against the limit, and the display logic was confirmed to match. Overspent categories now correctly show Over Budget and trigger the alert.
- **Verified by:** two automated regression tests.

BUG-10: Negative Amounts Displayed Incorrectly

- **Severity:** Medium
- **Where:** Currency formatting logic used across the app.
- **What happened:** Negative balances displayed a value that was off by a whole unit. A small negative amount would round to a noticeably larger negative figure than it should.
- **Cause:** The formatting logic divided negative numbers directly, and the way division rounds negative numbers in Python produced the error.
- **Fix:** The sign is now separated from the number before the calculation runs, then reattached afterward, producing the correct value.
- **Verified by:** an automated regression test.

4. Adaptive Maintenance: Date Range CSV Export

The CSV export previously produced every transaction in one file with no way to narrow it down. This phase added the option to export by date range: all transactions, the current month only, or a custom range chosen by the user. The interface shows a live count of matching transactions and names the exported file according to the range selected. This makes the export usable for monthly bookkeeping, tax season, or reconciling against a bank statement. The feature was verified with three automated tests, one for each export scope.

5. Perfective Maintenance: Bulk Delete

Deleting transactions previously required removing them one at a time. This phase introduced a multi select checklist, similar to an email inbox, including a Select All option, a preview of the selected transactions, and a two part confirmation step before anything is removed. A separate, clearly labeled option allows clearing the entire transaction table when needed. Behind the scenes, deletions are now handled as a single batch operation instead of one call per row, and budgets are automatically recalculated afterward. The feature was verified with two automated tests, covering both bulk deletion and full table clearing.

6. Complexity and Refactoring Metrics

The same complexity check used in Phase 4 was run again after this phase, to confirm the new features did not make the code harder to maintain:

Metric	Phase 4 Baseline	Phase 5 Now	What Changed
Functions analyzed	68	**73**	5 new functions added
Average cyclomatic complexity	3.07, Grade A	**3.01, Grade A**	Slightly simpler overall
Grade A functions (complexity 1 to 5)	60 (88.2%)	**65 (89.0%)**	5 more simple functions
Grade B functions (complexity 6 to 10)	5 (7.4%)	**5 (6.8%)**	Unchanged
Grade C functions (complexity 11 to 20)	3 (4.4%)	**3 (4.1%)**	Still limited to input validators
Maintainability index	Grade A	**Grade A**	Held across every module
Automated tests	144	**152**	8 new Phase 5 tests
Test pass rate and coverage	100% / 97%	**100% / 97%**	Zero regressions introduced

Result: the codebase got slightly simpler, even after adding new features.

7. Lehman's Laws of Software Evolution

A short review of ExpenseMate against M.M. Lehman's 8 laws, a set of observations about how long lived software systems behave over time:

Law	What It Means for ExpenseMate
I. Continuing Change	ExpenseMate operates in personal finance, where user needs keep shifting, so the system has to keep adapting. Moving from basic transaction handling to scoped exports and bulk actions is that adaptation in practice.
II. Increasing Complexity	Complexity normally creeps up as features are added. Splitting the new export and delete logic into small, focused functions actually brought average complexity down slightly, from 3.07 to 3.01.
III. Self Regulation	Defect counts fell steadily across phases: 5 in Phase 3, 3 in Phase 4, 2 in Phase 5, showing the process was maturing rather than destabilizing.
IV. Conservation of Organizational Stability	Both members kept a steady, comparable pace across phases even as the work shifted from building the architecture toward testing and maintenance.
V. Conservation of Familiarity	The bulk delete feature kept the old single item delete as the default and reused a familiar checkbox pattern, so users did not need to relearn anything.
VI. Continuing Growth	The feature set expanded every phase: from basic logging, to multi series analytics, to scoped exports and bulk management, which is expected for the system to stay useful.
VII. Declining Quality	Left unchecked, quality tends to slip over time. The two bugs fixed this phase are examples of that kind of decay, and the 152 test suite at 97% coverage is what keeps it in check.
VIII. Feedback System	A visible interface bug and friction in the old delete flow both directly shaped this phase's work, showing evolution here runs on real feedback rather than guesswork.

8. Conclusion

Phase 5 moved ExpenseMate through active maintenance without disrupting what already worked. Two real bugs were fixed, the export now adapts to how people actually use it, and bulk transaction management was added cleanly. Complexity metrics moved in the right direction (average cyclomatic complexity: 3.01, still Grade A), and all 152 tests pass at 97% coverage. Measured against Lehman's Laws, ExpenseMate is evolving the way a healthy system should: growing, staying disciplined about complexity, and responding to real feedback.